TRAN AN PHU

Ho Chi Minh City, Vietnam | anphusg@gmail.com

EDUCATION

Le Thanh Tong Primary, Secondary and High School

Ho Chi Minh City, Vietnam | Expected graduation: 2027

- **GPA:** Grade 9: 9.0/10; Grade 10: 9.0/10.

Score:

- IELTS 7.0
- AP Calculus AB 4, Calculus BC 5, Computer Science A 4, Statistics 4.

HONORS AND AWARDS

- Gold Medal, Southeast Asian Computer Science Olympiad (SEACSO), national round, 2025.
- Gold Medal, Silicon Valley International Invention Festival (SVIIF), for AI-Integrated Nutritional Care Software with Online Specialists, 2025.
- Gold Medal, 1 Idea 1 World, Istanbul, for Machine Learning-Based Safety Compliance Classification Using Camera Vision, May 2026.

RESEARCH

A Multi-View Anomaly Detection Framework for Elephant Movement Based on GPS Data

Co-author | June 2026 | The Indonesian Journal of Computer Science

DOI: 10.33022/ijcs.v15i3.5146

- Preprocessed 12,170 elephant GPS records and engineered 25 spatial, temporal, and movement features for anomaly analysis.
- Trained a compact Random Forest model using nine selected features, SMOTE, undersampling, and class weighting to address class imbalance.

TECHNOLOGY AND ENTREPRENEURSHIP PROJECTS

AI-Based Wildlife Monitoring Drone

Student Researcher and Prototype Developer | 2026

- Inspired by horseback riding and his observations of changes in horses' gait, explored how aerial footage and AI could help monitor animal movement.
- Designed and built a camera-equipped quadcopter prototype with onboard computing, flight-control, and GPS components.
- Collected video and field data on horses at a riding school and on wildlife, including elephants and gaur, in Mã Đà Forest.
- Annotated wildlife images in Roboflow, trained a YOLO26 object detector, and developed a workflow to pair image detections with UAV telemetry for ground-station review.

Machine Learning-Based Safety Compliance Classification Using Camera Vision

Model Development and Training | December 2025–May 2026

- Developed and trained the model that classifies camera images as Helmet or No Helmet to support workplace safety monitoring.
- The project received a Gold Medal at 1 Idea 1 World in Istanbul in May 2026.

AI-Integrated Nutritional Care Software with Online Specialists

Research Team Member, Database Management | April–August 2025

- Helped develop an AI-supported personalized nutrition platform with more than 350 EatClean recipes, nutrition data for more than 700 foods, and an online specialist chat feature.
- Applied data analysis and predictive modeling to support personalized recommendations; managed a MySQL database to organize nutrition data and improve data processing.
- The project received a Gold Medal at the 2025 Silicon Valley International Invention Festival in the United States.

TECHNOLOGY AND ENTREPRENEURSHIP PROJECTS CONTINUED

EduMath Lab

Founder and Developer | December 2025–Present

- Built an interactive learning platform for children ages 5–12 that combines mathematical reasoning, introductory programming, and algorithmic thinking through visual activities.
- Introduced the platform at Hi Vong School in District 6, Ho Chi Minh City, and provided 30 learning accounts to 30 students.
- Proposed next phase: offer accounts at US$10 per month and donate all subscription revenue to shelters in Ho Chi Minh City.

KidAssist

Software Developer | October 2024–September 2025

- Developed a mobile app for children with disabilities and their families, providing practical resources, parent-to-parent information sharing, and connections to specialists.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Tekite Computer Science Club

President, 20-Member Club | August 2024–Present

- Coordinated members, technology projects, and STEM outreach; initiated and ran a science fair for 50 primary school students in Núi Tượng, Đồng Nai, featuring water rockets, light refraction, and water-pump experiments.

SENTEM Đồng Tháp 2025 | *Project Lead and Volunteer Instructor* | Summer 2025

- Designed and taught workshops on water-rocket assembly and programming Arduino-based Bluetooth cars; organized rocket launches and car races for nearly 100 students.
- Helped raise and distribute 100 gift packages and 20 scholarships; the program reported a total value of VND 30 million.

Summer Computing and Engineering Outreach | *Tekite Organizing Team Member* | Summer 2026

- Taught Scratch game programming to 30 students facing financial hardship at Hi Vong School in District 6, Ho Chi Minh City.
- Worked with three craft and science clubs on a camp for 200 ethnic-minority primary students at Vo Thi Sau Primary School in Dak Lak Province; Tekite led model race-car assembly and a remote-controlled obstacle-course race.
- The joint program distributed 200 learning-supply packages and served a buffet meal, with a reported total value of VND 50 million.

Green Defender

President | July 2025–Present

- Partnered with Cat Tien National Park on a conservation learning visit for 20 members covering forest ecosystems, wildlife, hydroponics, and supervised animal-care activities.
- Worked with the Đồng Nai Nature and Culture Reserve Management Board on a habitat-restoration program that planted 300 trees; the reported project value was VND 30 million.

TECHNOLOGY TRAINING

Ho Chi Minh City University of Technology

IoT Program Participant | July–August 2024

- Studied IoT and helped develop a radar-sensor obstacle-avoiding smart car and a smart-home project.

Saigon Hi-Tech Park

Technology Program Participant | June–July 2025

- Practiced robotics, AR/VR, AI, and 3D scanning; studied industrial production processes and assessed potential market applications and product ideas.
- Developed and presented an idea to use AI to inspect drill-bit quality for ANMI.

CULTURAL PROJECT

Flow of the Folk

Water Puppet Performer and Organizer | August 2024

- Studied Vietnamese water puppetry and performed at a fundraising event for 50 attendees at Phương Nam Water Puppet Theater; the event raised VND 20 million for Tâm Đức Shelter.
- Helped introduce water puppetry to children at Tâm Đức Shelter through a cultural exchange.